

Statement of Work

Prefabricated Cured Propellant Test Facility (Armag Corporation)

1.0 Scope

The purpose of this Statement of Work (SOW) is to define the requirements for the procurement, fabrication, delivery, and installation of a **prefabricated Hazard Division (HD) 1.3 cured propellant test facility** to support the mission needs of the Solid Propulsion and Pyrotechnic Devices Branch (ER62). The required facility is a **30' × 15' × 10' explosion-resistant operations building**, constructed to ATF Type 2 magazine specifications and equipped with explosion-resistant power, lighting, HVAC, grounding, ESD flooring, and non-sparking interior laboratory fixtures. The building configuration shall be based on the engineering drawings.

2.0 Applicable Documents

The contractor shall comply with the following documents as applicable:

- ATF Specification for Type 2 Explosives Storage Magazines (27 CFR §555.208)
 - Armag Corporation Drawings and Bill of Materials for unit S4276 ([S4276](#)).
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3.0 Requirements

3.1 General Requirements

The contractor shall design, fabricate, and deliver a fully assembled prefabricated operations building meeting the following requirements:

- Dimensions: **30 ft (L) × 15 ft (W) × 10 ft (H)**.
- Structure shall meet or exceed ATF Type 2 magazine construction requirements.
- Exterior finish: Desert Tan Devthane 349QC, consistent with [S4276](#).

- All systems shall be intrinsically safe or explosion resistant per Class II Div 1 requirements.

3.2 Electrical System Requirements

The building shall include an integrated explosion-resistant power distribution system as shown in [S4276](#)

- 120/208V 3-phase main panel, NEMA 3R.
- 100A main breaker.
- 50A breaker and transformer for oven/480V outlet.
- Multiple 120V/20A explosion-resistant receptacles.
- Explosion-resistant LED lighting (45-Watt), minimum three fixtures.
- Explosion-resistant exit light.
- Explosion-resistant camera receptacles as quoted.

3.3 HVAC Requirements

The contractor shall provide:

- **Class II, Division 1 explosion-resistant HVAC unit**, 36,000 BTU rating (Scientific Systems HC-WACX-3503 or equivalent).
- **Class II, Division 1 explosion-resistant heater** (Ruffneck FX6-208360-050-T or equivalent).

3.4 Grounding and Lightning Protection

The facility shall include:

- Copper bus bar grounding system and isolated ground plates.
- Lightning protection kit with NFPA-compliant ground rods.

3.5 Flooring and Surfaces

- ESD-rated conductive tile flooring (United Static Control “Titanium Dark Gray”).
- 2" insulation and 3" hardwood floor layer below tile.
- Painted plywood interior wall finish.

3.6 Laboratory Fixtures

- Aluminum non-sparking work benches (10'×3' and 5'×3'), shelves, and isolated-ground workstations.
- Self-contained eye-wash station.
- Floor drain.

3.7 Penetrations and Interfaces

The building shall include:

- Through-wall steel pipe penetrations (up to 4" diameter) for oven exhaust and instrument cabling.
- Exterior NEMA 3R disconnect for 480V oven power.

3.8 Doors and Security

- High-security double exterior doors with ATF-compliant hooded locking systems.
- Panic bar on active door.

3.9 Delivery Requirements

- Delivery FOB Destination: MSFC, Huntsville, AL
- Contractor shall coordinate with MSFC for delivery routing and off-loading requirements.
- Estimated availability: approximately 30 weeks after drawing approval.

4.0 Government-Furnished Property (If Applicable)

At this time, MSFC does not anticipate providing Government-Furnished Property (GFP). Any GFP will be identified in writing if required later.

5.0 Inspection and Acceptance Criteria

Acceptance shall be based on:

1. Review of as-built drawings and documentation package.
2. Verification that all electrical, mechanical, and safety systems match specifications in [S4276](#)

3. Visual inspection upon delivery for damage or deviation.
 4. Functional demonstration of:
 - HVAC operation
 - Lighting and electrical outlets
 - Grounding and continuity checks
 - Door and security hardware operation
 5. Any discrepancies must be corrected prior to final acceptance.
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6.0 Warranty

The contractor shall provide warranty coverage consistent with the terms described in including:

- 12-month warranty on materials and workmanship.
 - 12-month paint warranty against flaking/peeling.
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7.0 Points of Contact

NASA/MSFC Technical POC: